Academic Restructuring Proposal

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Thinking ● Planning academic comparisons I’m interpreting “universe” …

Thinking

- **Planning academic comparisons**

  I'm interpreting "universe" as space science and engineering, including satellites, Earth observation, atmosphere, and geospatial information. I'll compare university structures, curricula, timing, methods, and interdisciplinary models.

- **Comparing organizational models**

  I'm expanding the comparison across research-intensive and specialized universities. The key design question is whether this should be one broad school or a matrix of focused departments connected through shared programs and laboratories.

- **Compared academic organizational models**

- **Gathering official evidence**

  I'm checking institutional pages and curriculum information for representative universities, focusing on how geology, engineering, space science, atmosphere, energy, resources, and geospatial fields are organized and taught.

- **Gathered official evidence**

- **Refining the framework**

  I'm separating fundamental space science from space engineering and distinguishing core degrees from specialized tracks. I'm also testing whether tourism development belongs under spatial planning rather than geoscience.

- **Structuring the faculty**

  I'm converging on a matrix model: focused departments for geoscience, resources, civil engineering, and geospatial-space technologies, connected through shared programs. Tourism development fits best within spatial planning or geotourism, not as a core geoscience discipline.

- **Designing the academic structure**

  I'm considering four departments: Earth and Resource Sciences; Mineral and Energy Engineering; Civil and Geological Engineering; and Space, Remote Sensing, and Geospatial Engineering. This preserves disciplinary depth while supporting shared interdisciplinary programs.

- **Mapping programs and facilities**

  I'm aligning the structure with three pillars while keeping four departments. Shared laboratories should support multiple programs, while curricula should combine

common foundations, disciplinary sequences, fieldwork, internships, and capstone projects.

- **Specifying curricula**

  I'm organizing majors around Earth sciences, resource engineering, civil/geotechnical engineering, and space-geospatial technology. Each pathway needs clear prerequisites, practical training, fieldwork, internships, and a final interdisciplinary design or research project.

- **Integrating course design**

  I'm selecting interdisciplinary courses that connect geology, infrastructure, geospatial data, and space applications. Examples include Earth observation, subsurface modeling, hazards, resilient infrastructure, planetary geology, digital twins, and an integrated design capstone.

- **Refined academic framework, structure, programs, facilities, curricula, and courses**

- **Synthesizing the framework**

  I'm consolidating the research into a comparative structure: international models, a recommended four-department organization, curriculum pathways, shared facilities, interdisciplinary courses, and implementation priorities for a coherent faculty.

- **Synthesized the framework**

- **Finalizing program architecture**

  I'm resolving the detailed indexes as core degree programs plus specialization tracks, avoiding too many standalone majors. Tourism development becomes a geospatial-planning or geotourism track, while drilling remains a shared engineering platform.

- **Finalized program architecture**

- **Finalizing degree pathways**

  I'm retaining nine core programs: geology and mineral exploration, geophysics, geochemistry, geological and drilling engineering, mining and mineral processing, petroleum engineering, civil and geotechnical engineering, space science, and geomatics. Geotourism remains a specialization.

- **Clarifying program boundaries**

  I'm distinguishing space science, space engineering, atmospheric science, and geospatial technology rather than combining them loosely. I'm also placing engineering investigation under geological/geotechnical engineering and treating geotourism as a planning specialization.

- **Designing the four-year model**

I'm organizing eight semesters around shared foundations, disciplinary cores, specialization, fieldwork, internships, and a capstone. I'll use proportions rather than fixed credits because national credit systems differ.

- **Reconciling academic boundaries**

  I'm separating departments from shared laboratories: departments own curricula, while laboratories support multiple programs. A three-school model can contain seven departments, preserving disciplinary depth while enabling common courses, fieldwork, and integrated capstones.

- **Structuring the faculty**

  I'm organizing the faculty around four departments: Earth Sciences and Exploration; Mineral and Energy Engineering; Civil and Geological Engineering; and Space, Geomatics, and Earth Observation. Existing laboratories should support multiple programs, not define academic boundaries.

- **Structuring the faculty**

  I'm organizing the faculty around four departments, shared laboratories, and eight major programs. I'm keeping geotourism and regional planning as an interdisciplinary concentration rather than a standalone engineering degree.

- **Structuring academic programs**

  I'm organizing the faculty around four departments, shared laboratories, and focused majors. I'm separating geoscience, resource engineering, civil infrastructure, space, and geospatial studies while treating geotourism as an interdisciplinary concentration.

- **Structuring the faculty**

  I'm converging on five departments: Earth Sciences and Mineral Exploration; Applied Geophysics and Geochemistry; Mineral and Energy Engineering; Civil and Geological Engineering; and Space Science, Geomatics, and Earth Observation. This separates academic ownership from shared laboratories and interdisciplinary centers.

- **Structuring academic units**

  I'm organizing the faculty around five departments and shared research platforms. This keeps geology, exploration, engineering, civil infrastructure, space, geospatial studies, drilling, and regional development connected without duplicating responsibilities.

- **Structured faculty, academic units, and four-year degree pathways**

- **Structuring the faculty**

  I'm converging on five departments supported by shared laboratories: Earth Sciences and Mineral Exploration; Applied Geophysics and Geochemistry; Mineral and Energy Engineering; Civil and Geological Engineering; and Space Science and Geospatial Technology.

- **Clarifying the structure**

I'm organizing the proposal around five departments, shared research platforms, and eight specialized majors. This keeps laboratories collaborative while giving each degree program clear academic ownership and avoids unnecessary duplication.

- **Structuring academic programs**

  I'm organizing nine professional majors under five departments, while keeping geotourism and regional development as an interdisciplinary track. This separates scientific exploration, engineering, civil infrastructure, space science, and geospatial technology clearly.

- **Structuring academic programs**

  I'm organizing the faculty around five departments, nine professional majors, shared laboratories, and a four-year progression from foundations to specialization, practice, and capstone work. Geological, engineering, space, and geospatial subjects remain distinct but connected.

- **Structuring academic programs**

  I'm organizing the faculty around five departments, nine professional majors, shared research centers, and cross-disciplinary courses. I'm also keeping the requested four-year, eight-semester model while separating academic ownership from laboratory management.

- **Designing the academic structure**

  I'm consolidating overlapping units into five departments, nine majors, and shared research centers. I'll separate geology from engineering, keep space disciplines distinct, and treat geotourism as an interdisciplinary concentration.

- **Designed the faculty and academic program structure**

- **Designing the faculty structure**

  I'm organizing the proposal around five departments, nine major programs, shared laboratories, and interdisciplinary tracks. The main goal is reducing duplication while preserving strong science, engineering, geospatial, and space capabilities.

- **Designed the faculty structure**

- **Searched 48 websites**

✓ Worked for 3m 41s

Strategic academic restructuring proposal for the Faculty of Geoscience and Technology

First, congratulations on your election as Dean. You have an important opportunity to establish a clear academic identity for the faculty and develop an educational system

that responds to the future needs of the mineral resources, civil infrastructure, petroleum, and space industries.

Having considered your description, I believe your basic direction is correct: **the Faculty of Geoscience and Technology should retain both theoretical geoscience and applied engineering as its principal academic pillars.**

However, I would recommend a more systematic approach than simply reorganizing the existing laboratories or adding more specialized majors.

The faculty should be designed around three academic missions:

1. **Understanding the Earth:** geological sciences, mineral resources, petroleum geology, geophysics, geochemistry, hydrogeology, and Earth systems.
2. **Developing and utilizing the Earth:** mineral exploration, mining engineering, petroleum engineering, civil and geological engineering, drilling, and mineral materials.
3. **Understanding and utilizing space and geospatial technology:** space science, satellite engineering, Earth observation, remote sensing, surveying, and geospatial information systems.

These missions should be supported by a common foundation in mathematics, physics, chemistry, computing, and engineering sciences.

One important terminology clarification: I will interpret your term *universe section* as **Space Science and Space Engineering**, including satellite technology, Earth observation, atmospheric science, and geospatial applications. I will also interpret *exporters* as *experts or highly qualified specialists*.

My central recommendation is to establish a faculty with **three academic schools or departments, supported by interdisciplinary research laboratories and shared educational facilities**. This would preserve the integrated nature of your faculty without sacrificing academic specialization.

1. Is your current academic organization system suitable?

1.1 International academic practice: what do leading universities do?

I examined the organizational structures and educational programs of universities in North America, Europe, China, and Japan. An important finding is that there is no

single international organizational model.

Instead, three patterns are evident:

- **Disciplinary separation:** theoretical Earth sciences, civil engineering, and space engineering are organized in separate departments or faculties.
- **Integrated geoscience:** geology, geophysics, geochemistry, remote sensing, and space science are combined within one academic school.
- **Integrated science and engineering:** geoscience and engineering are brought together under a common institutional mission, while specialized departments retain responsibility for their respective disciplines.

The following examples are particularly relevant to your faculty.

1.2 Comparison of international universities

The comparison focuses on academic organization and undergraduate education, rather than university rankings.

University	Academic organization	Important lesson
Massachusetts Institute of Technology (MIT)	Earth, Atmospheric and Planetary Sciences (EAPS) and Civil and Environmental Engineering (CEE) are separate departments.	Separate disciplinary expertise can be connected through joint degrees.
University of Cambridge	Earth Sciences is taught within the Natural Sciences Tripos; Engineering has its own four-year program.	Strong common scientific education followed by specialization.
University of Oxford	Earth Sciences and Engineering Science are separate academic areas.	Scientific and engineering education can maintain distinct identities.
ETH Zurich	Earth and Planetary Sciences (D-EAPS) and Civil, Environmental and Geomatic Engineering (D-BAUG) are separate departments.	Geoscience, civil engineering, and geomatics can remain distinct but closely connected.
Peking University	The School of Earth and Space Sciences combines geology, geochemistry, geophysics,	An integrated Earth and Space academic school is a

University	Academic organization	Important lesson
	space science and technology, and geographic information science.	successful organizational option.
Jilin University	Earth Sciences, Geoexploration Science and Technology, and Construction Engineering are organized in separate colleges.	Different technical disciplines can be coordinated through specialized academic units.
University of Tokyo	Earth and Planetary Sciences are in the Faculty of Science; Civil Engineering, Systems Innovation and Aeronautics and Astronautics are in Engineering.	Clear disciplinary specialization with interdisciplinary engineering education.
Colorado School of Mines	Earth, Space and Planets and Minerals and Subsurface Energy are organized as separate schools within one specialized university.	A particularly useful model for connecting theoretical geoscience with resource engineering.

Sources: MIT's EAPS and joint-major programs [Undergraduate Program; First-Year Undergraduates]; Cambridge and Oxford course structures [Undergraduate; Course overview; Course Information]; ETH departmental structure [Departments; Study Earth, Climate, and Space Sciences – Department of Earth and Planetary Sciences]; Peking University [专业介绍-北京大学地球与空间科学学院; 学院简介-北京大学地球与空间科学学院]; Jilin University [Education-地球探测科学与技术学院; Introduction-地球科学学院英文]; University of Tokyo [Faculty of Science; (4) Undergraduate Departments]; Colorado School of Mines [Programs; Geophysics].

1.3 Particularly important examples

A. Peking University: integrated geoscience and space science



Peking University's School of Earth and Space Sciences is especially relevant to your proposed structure.

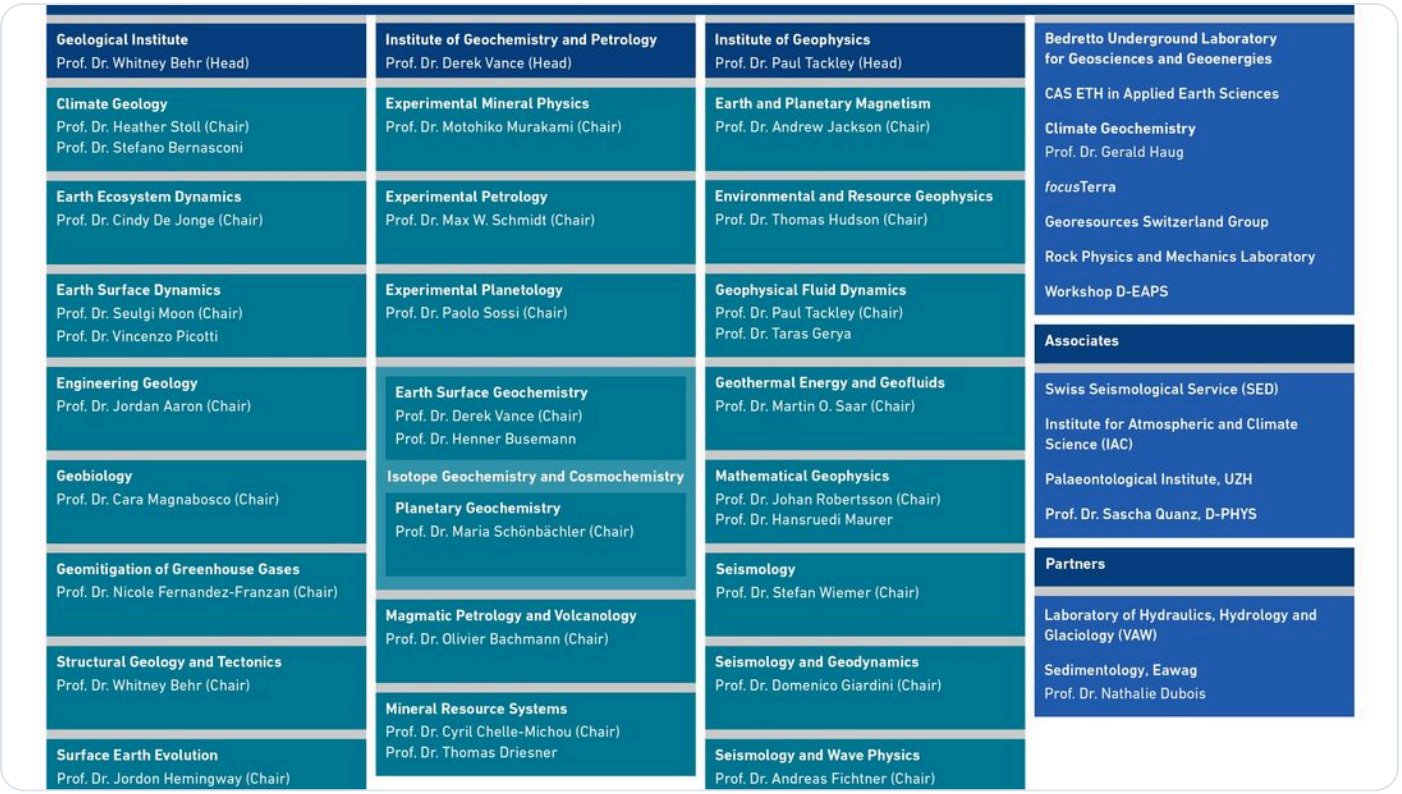
It combines five undergraduate disciplines:

1. Geology
2. Geochemistry
3. Geophysics
4. Space Science and Technology
5. Geographic Information Science

Students first study mathematics, physics, chemistry, and computing, before choosing their specialized discipline.

This is a very good model for integrating your theoretical geoscience and space science programs. However, civil engineering is not part of this particular school. [专业介绍-北京大学地球与空间科学学院; 培养方案-北京大学地球与空间科学学院]

B. ETH Zurich: clear boundaries and interdisciplinary collaboration



ETH Zurich has a clear division between:

- Department of Earth and Planetary Sciences.
- Department of Civil, Environmental and Geomatic Engineering.

Nevertheless, its Space Systems master's program combines aerospace engineering, planetary science, Earth observation, robotics, and space communication.

Its Geospatial Engineering bachelor's program also integrates mathematics, computing, geodesy, measurement technology, GIS, cartography, satellite navigation, and spatial planning.

This is an excellent example of how academic boundaries can be clear while educational programs remain interdisciplinary. [Departments; Master Space Systems; Bachelor Geospatial Engineering]

C. Colorado School of Mines: science and engineering under one institutional mission



This university is probably the most relevant applied-science example for your faculty's resource development objectives.

Its School of Earth, Space and Planets offers geological engineering and geophysical engineering. Its School of Minerals and Subsurface Energy offers mining engineering and petroleum engineering.

It also connects resource exploration with mineral processing, GIS, geoinformatics, space resources, and sustainable resource development.

The arrangement demonstrates that theoretical geoscience and applied engineering can work under one institutional mission without being forced into the same academic department. [Programs; Programs; Geophysical Engineering Major]

D. Jilin University: a useful organizational reference for your existing system



Jilin University has a particularly interesting structure.

Its College of Geoexploration Science and Technology contains teaching departments covering:

- Applied Geophysics
- Planetary and Solid Earth Geophysics
- Geochemistry
- Surveying and Mapping Engineering
- Remote Sensing and Geographic Information Systems

Its undergraduate programs include Geophysics, Exploration Technology and Engineering, Surveying and Mapping Engineering, and Geographic Information Science.

Separately, its Earth Sciences college covers geology, resource engineering, energy sciences, and land and resources. Its Construction Engineering college covers geological engineering and civil engineering.

This resembles the academic diversity of your existing faculty, but the disciplines have clearer organizational responsibilities. [学院概况-地球探测科学与技术研究; Education-地球探测科学与技术学院; Introduction-地球科学学院英文; 吉林大学建设工程学院2026年本科招生专业介绍-吉林大学建设工程学院]

1.4 My assessment of your present system

Strategic direction is appropriate

I would not recommend abandoning your integrated faculty concept.

Your current system has several important advantages:

- It can educate specialists who understand the relationship between geological conditions and engineering solutions.
- It can combine mineral exploration, drilling, mining, and mineral processing.
- It can connect civil engineering with geological investigation and geotechnical conditions.
- It can connect space technology with remote sensing, surveying, and geological information.
- It can develop interdisciplinary research and industry partnerships.

However, your present organization has three structural weaknesses.

Weakness 1: Academic disciplines and research laboratories are mixed together.

For example, mineralogy and petrology are fundamental geoscience disciplines, whereas mineral material development and rock application are engineering and technology disciplines. They are related, but they do not necessarily belong to the same academic department.

Similarly, electromagnetic prospecting, magnetic prospecting, and prospecting equipment development are different research and technical capabilities, rather than three independent academic disciplines.

Weakness 2: Some major academic disciplines are missing or insufficiently represented.

In particular, your faculty should strengthen:

- Economic geology and mineral deposit modeling.
- Mineral resource estimation and evaluation.
- Hydrogeology and groundwater engineering.
- Geological engineering and geotechnical engineering.

- Mining engineering and mine planning.
- Mineral processing and extractive metallurgy.
- Petroleum reservoir engineering and production engineering.
- Satellite remote sensing and geospatial data science.
- Structural engineering and construction management.

Weakness 3: Your major indexes do not yet form a sufficiently clear educational hierarchy.

For example, mineral exploration, geophysical exploration, and geochemical exploration are closely related. Students should receive a common foundation and then specialize in these areas.

Likewise, surveying, GIS, and remote sensing should share common foundational subjects rather than repeatedly teaching similar material in different major programs.

My conclusion: Your faculty should be an integrated science-and-engineering faculty, but its internal academic organization should be based on clearly defined disciplines, professional responsibilities, and research fields.

This is consistent with the international examples above. The important distinction is that *integration at the faculty level does not require the elimination of specialization at the department level.*

1.5 Additional reference: Tsinghua University and UC Berkeley

Two further examples help clarify the boundary between academic disciplines.

- Tsinghua University maintains a School of Civil Engineering, a School of Aerospace Engineering, and Earth System Science under its School of Sciences. Its undergraduate degrees are generally four years. This is a good example of clearly separated scientific and engineering disciplines. [Degree Programs-Tsinghua University; Schools & Departments-Tsinghua University; Tsinghua University offers a new undergraduate major in earth system science-Department of Earth System Science,Tsinghua University]
- University of California, Berkeley has a GeoSystems Engineering course pathway within Civil and Environmental Engineering. It includes engineering geology, applied geophysics, groundwater, geomatics, and geotechnical engineering. This

demonstrates that important interdisciplinary subjects can be taught across traditional departmental boundaries. [Courses; CEE]

The international comparison suggests the following principle:

Organize academic disciplines according to their fundamental scientific and engineering identities, but organize education and research projects according to the problems they solve.

This is the principle I recommend adopting.

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2. Recommended academic organization system for your faculty

2.1 Proposed organizational structure

I recommend establishing **five academic departments within the Faculty of Geoscience and Technology**.

This structure is designed to preserve the scientific and engineering character of your faculty while creating clearer academic responsibilities.

FACULTY OF GEOSCIENCE AND TECHNOLOGY

Dean · Faculty Academic Council · Quality Assurance Committee

1. Department of Earth Sciences and Mineral Exploration

Geology, mineral deposits, petroleum geology, hydrogeology and resource evaluation.

2. Department of Applied Geophysics and Geochemistry

Geophysical and geochemical exploration, instrumentation, data processing and interpretation.

3. Department of Mineral, Mining, Petroleum and Drilling Engineering

Mining, mineral processing, petroleum production, drilling and subsurface engineering.

4. Department of Civil and Geological Engineering

Structural engineering, geotechnical engineering, construction, infrastructure and engineering geology.

5. Department of Space Science and Geospatial Technology

Space science, satellite technology, atmospheric science, planetary science, surveying, remote sensing, GIS and geospatial information.

Shared research and educational facilities

Six laboratories and one faculty-wide drilling and subsurface engineering center. These facilities serve students and researchers from multiple departments.

The five departments should have their own academic responsibilities, but they should not become isolated from one another.

2.2 Responsibilities of the five departments

Department 1. Earth Sciences and Mineral Exploration

Its main responsibility is to understand the geological structure, composition, evolution, and resource potential of the Earth.

Recommended academic divisions:

1. General and Structural Geology
2. Mineralogy and Petrology
3. Economic Geology and Ore Deposit Geology
4. Mineral Exploration Geology and Resource Evaluation
5. Petroleum Geology and Basin Analysis
6. Hydrogeology and Groundwater Science

This department should be responsible for the theoretical foundations of mineral and petroleum exploration.

For example, it should teach students how mineral deposits form, how geological structures control ore bodies, how to recognize geological indicators, and how to estimate the geological potential of mineral resources.

Department 2. Applied Geophysics and Geochemistry

Its main responsibility is to develop and apply physical and chemical methods for detecting underground resources and geological structures.

Recommended academic divisions:

1. Seismic and Seismic Exploration
2. Gravity and Magnetic Exploration

3. Electrical and Electromagnetic Exploration
4. Exploration and Environmental Geochemistry
5. Geophysical and Geochemical Instrumentation
6. Exploration Data Processing, Inversion and Interpretation

This department should connect fundamental physics and chemistry with practical mineral and petroleum exploration.

For example, students should learn how to design geophysical surveys, select appropriate prospecting methods, acquire field data, process and interpret the data, and integrate the results with geological information.

Jilin University's exploration technology curriculum is a useful reference: it combines gravity, magnetic, electrical, seismic and borehole geophysics with geochemical analysis and data interpretation. [Exploration Technology and Engineering-地球探测科学与技术学院; Geophysics-地球探测科学与技术学院]

Department 3. Mineral, Mining, Petroleum and Drilling Engineering

Its main responsibility is the engineering development, extraction, processing and utilization of underground resources.

Recommended academic divisions:

1. Mining Engineering
2. Mineral Processing Engineering
3. Extractive Metallurgy and Mineral Materials Engineering
4. Petroleum Engineering
5. Drilling and Subsurface Engineering
6. Mining and Petroleum Machinery and Automation

This department should emphasize engineering design, operational technology, production, safety, economics and environmental management.

The separation between geology and engineering is important here.

For example:

- Geology determines the location, geological characteristics and potential quantity of an ore deposit.
- Applied geophysics and geochemistry help identify and delineate the deposit.
- Drilling engineering obtains subsurface information and supports resource development.
- Mining engineering designs and operates the mine.
- Mineral processing engineering separates and concentrates valuable minerals.

These disciplines should cooperate in the same resource-development projects.

Department 4. Civil and Geological Engineering

Its main responsibility is the investigation, design, construction and management of civil infrastructure.

Recommended academic divisions:

1. Structural and Building Engineering
2. Geotechnical Engineering and Foundation Engineering
3. Engineering Geology and Geological Investigation
4. Underground and Tunnel Engineering
5. Transportation and Infrastructure Engineering
6. Hydraulic and Water Resources Engineering
7. Construction Engineering and Management

The combination of geological engineering and civil engineering is particularly important.

Geological conditions influence the design of foundations, tunnels, dams, bridges, roads, underground structures and other infrastructure.

Tsinghua's civil engineering program, for example, includes structures, underground structures, bridge and road engineering, earthquake resistance, and disaster prevention. [Undergraduate-School of Civil Engineering Tsinghua University; Degree Programs-Tsinghua University]

Department 5. Space Science and Geospatial Technology

Its main responsibility is to study the space environment, develop space-related technologies, and obtain and utilize geospatial information.

Recommended academic divisions:

1. Space Physics and Space Environment
2. Atmospheric Science and Earth Observation
3. Planetary Science and Planetary Exploration
4. Satellite and Space Systems Engineering
5. Surveying, Geodesy and Satellite Navigation
6. Remote Sensing and Geographic Information Systems
7. Geospatial Data Science and Spatial Planning

I recommend distinguishing three areas within this department:

- Space science: understanding the physical processes of space, planetary environments and the atmosphere.
- Space engineering: developing satellite systems, sensors, instruments and space missions.
- Geospatial engineering: collecting, processing and applying spatial information using surveying, GNSS, GIS and remote sensing.

They are closely related but are not identical disciplines.

MIT and ETH Zurich provide good examples of space engineering education through subjects such as satellite engineering, orbital dynamics, space systems engineering, planetary science and Earth observation. [Department of Aeronautics and Astronautics; Education - ETH Zurich]

If your faculty does not yet have sufficient satellite engineering equipment and specialists, I would initially emphasize Space Science, Earth Observation and Geospatial Technology, and develop a full Space Systems Engineering program progressively.

2.3 How should your existing laboratories be reorganized?

My recommendation is to distinguish between **academic departments and research laboratories**.

A department should be responsible for education, academic discipline development, teaching staff and degree programs.

A laboratory should be responsible for research equipment, experiments, field measurements, analytical methods and practical training.

I would retain your six laboratories, but revise their academic boundaries as follows.

Existing laboratory	Recommended role and organization
Resource Exploration Laboratory	Geological mapping, mineral deposits, mineral exploration, analytical geochemistry and resource evaluation.
Civil Engineering Laboratory	Structural engineering, construction, geotechnical investigation, foundation engineering and infrastructure design.
Mineral Engineering Laboratory	Mineral processing, extractive metallurgy, mineral materials and rock utilization.
Geophysical Prospecting Laboratory	Seismic, gravity, magnetic, electrical, electromagnetic methods, instruments and data processing.
Petroleum Laboratory	Petroleum geology support, drilling, reservoir engineering, production, machinery and petroleum processing.
Space Laboratory	Space science, atmospheric science, satellite data, remote sensing, surveying and geospatial information.

In addition, I recommend establishing a:

Faculty-wide Drilling and Subsurface Engineering Center

This center should coordinate drilling technology, drilling equipment, borehole measurements, drilling fluids and subsurface investigation.

It should serve the geological exploration, mining, petroleum and civil engineering departments.

Your existing Drill Engineering Division should have its academic home in Department 3, while the center itself should serve all five departments.

There are also several divisions whose academic responsibilities should be adjusted:

- Mineralogy and Petrology should be academically assigned to Department 1, although the laboratory facilities may remain shared with Department 3.
- Geochemical Prospecting should be assigned to Department 2.
- Petroleum Prospecting should be academically assigned to Department 1.
- Petroleum Mining Engineering, Petroleum Machinery and Petroleum Processing should be assigned to Department 3.
- Tourist District Development should be developed as an interdisciplinary program involving Department 4 and Department 5.
- Surveying should be organized within Department 5, with engineering surveying courses shared with Department 4.

This arrangement will reduce overlapping responsibilities and make academic cooperation easier.

2.4 How to make cooperation between departments flexible

I recommend creating three faculty-wide academic mechanisms.

1. Faculty Academic and Curriculum Committee

Responsibilities:

- Establish common educational standards.
- Approve the curriculum of each major.
- Eliminate overlapping subjects.
- Coordinate common courses between departments.
- Review course content every academic year.
- Evaluate graduate competencies.

2. Interdisciplinary Research and Education Committee

This committee should organize research and educational projects involving several departments.

For example, a mineral exploration project could involve:

- Department 1: geological interpretation.

- Department 2: geophysical and geochemical surveys.
- Department 3: drilling and mineral development.
- Department 5: satellite remote sensing and GIS.

3. Industry Advisory Committee

Invite experts from geological surveys, mining companies, petroleum companies, construction companies, satellite and geospatial companies, and government technical institutions.

They should participate in:

- Reviewing curriculum content.
- Identifying new technical requirements.
- Providing internships.
- Supervising student projects.
- Evaluating graduates' practical capabilities.

This is particularly important because your faculty aims to produce internationally competitive specialists, rather than graduates who possess only theoretical knowledge.

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3. Recommended detailed major indexes

I recommend organizing the major programs into three broad academic areas, with nine principal professional majors.

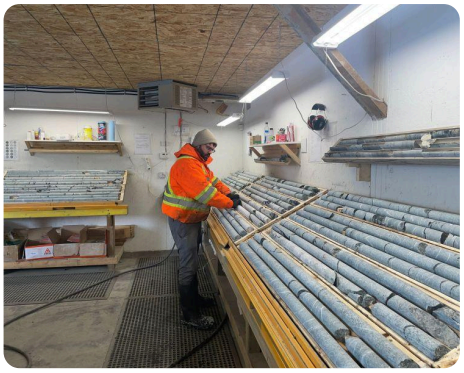
The following structure covers your existing detailed major indexes while adding some important disciplines that are currently insufficiently represented.

3.1 Geoscience and exploration section












No.	Recommended major	Main specialization areas
G1	Geology and Mineral Exploration	General geology, structural geology, economic geology, ore deposits, mineral exploration, resource estimation
G2	Applied Geophysics	Seismic, gravity, magnetic, electrical, electromagnetic and borehole geophysics
G3	Applied Geochemistry	Exploration geochemistry, analytical geochemistry, petroleum geochemistry and environmental geochemistry
G4	Petroleum Geoscience	Petroleum geology, sedimentology, basin analysis, reservoir geology and oil and gas exploration
G5	Hydrogeology and Environmental Geoscience	Groundwater exploration, hydrogeology, environmental geology, water-resource assessment and geological hazards

I would give particular attention to G1, G2 and G3 because these form the core of mineral exploration.

The distinction between G1, G2 and G3 should be maintained:

- G1 studies geological conditions and mineral deposit formation.

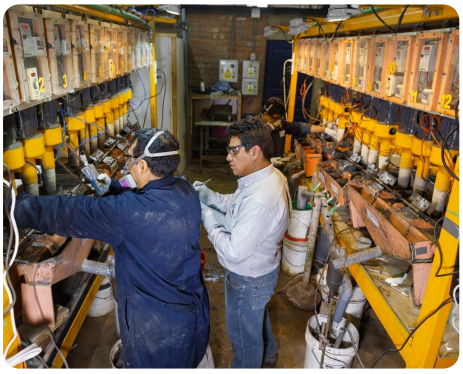
- G2 applies physical measurements to investigate underground structures.
- G3 applies chemical measurements to identify geochemical anomalies and mineralization.

However, their students should undertake integrated exploration projects.

G4 and G5 may initially be offered as specializations within G1 if the faculty lacks sufficient staff to establish independent degree programs.

3.2 Engineering and technology section







No.	Recommended major	Main specialization areas
E1	Geological Engineering and Drilling Engineering	Geological investigation, drilling technology, borehole engineering, geotechnical investigation, deep drilling

No.	Recommended major	Main specialization areas
E2	Mining and Mineral Processing Engineering	Mine design, mine planning, underground mining, open-pit mining, mineral processing, extractive metallurgy
E3	Petroleum Engineering	Drilling and completion, reservoir engineering, petroleum production, well testing, petroleum facilities and processing
E4	Civil and Geotechnical Engineering	Structural engineering, building engineering, foundation engineering, underground construction, transportation and water infrastructure

I would particularly emphasize E1 and E2 because they are closely related to your faculty's mineral exploration objectives.

E1 should provide students with the capability to design and operate drilling systems for mineral exploration, petroleum exploration and geological investigation.

E2 should cover the complete engineering chain from mine planning to mineral processing and the production of valuable mineral materials.

Colorado School of Mines provides useful examples of mining and petroleum curricula. Its mining program includes mine design, rock mechanics and mineral processing, while its petroleum program includes drilling, reservoir engineering, completion engineering, production, economics and data analytics. [Bachelor of Science in Mining Engineering; Bachelor of Science in Mining Engineering; Bachelor of Science in Petroleum Engineering]

For civil engineering, I would not create too many narrowly divided majors. Instead, use E4 as the main degree program with four specializations:

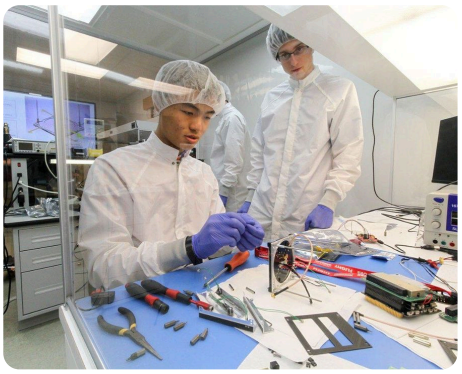
1. Structural and Building Engineering
2. Geotechnical and Underground Engineering
3. Transportation and Infrastructure Engineering
4. Hydraulic and Water Resources Engineering

This will allow your civil engineering program to maintain a sufficient common professional foundation.

3.3 Space and geospatial section









No.	Recommended major	Main specialization areas
S1	Space Science and Technology	Space physics, planetary science, atmospheric science, space weather and space exploration
S2	Space Systems and Satellite Engineering	Satellite design, orbital mechanics, spacecraft instrumentation, satellite communication and mission engineering
S3	Geomatics and Geospatial Engineering	Surveying, geodesy, GNSS, photogrammetry, GIS, remote sensing and spatial data analysis

S2 could initially be a specialization within S1 rather than a separate undergraduate degree.

The important point is that your Space Laboratory should not be restricted to theoretical space science. It should gradually develop applied capabilities in satellite systems, Earth observation and geospatial technologies.

Peking University's Space Science and Technology program provides a reference for the scientific side, including space plasma physics, magnetospheric physics, atmospheric physics and space weather. MIT and ETH Zurich provide useful examples of the engineering side, including orbital dynamics, satellite systems and spacecraft design. [SCHOOL OF EARTH AND SPACE SCIENCE, Peking University; Department of Aeronautics and Astronautics; Master Space Systems]

3.4 Where should Tourist District Development belong?

I would not recommend establishing Tourist District Development as an independent engineering major within the faculty at this stage.

Instead, establish an interdisciplinary specialization named:

Geotourism, Geoheritage and Regional Development

It could be offered under S3, with support from E4 and the geology department.

Recommended subjects:

- Geological Heritage and Geopark Development
- Geomorphology and Landscape Evaluation
- Geographic Information Systems for Tourism Planning
- Regional and Spatial Planning
- Sustainable Tourism Infrastructure
- Environmental Impact Assessment
- Tourism Resource Evaluation and Development

This specialization would train students to identify geological and natural attractions, evaluate their development potential, plan infrastructure, and support sustainable regional tourism.

3.5 Final recommended major structure

For clarity, I would use the following hierarchy:

Three academic areas

Area A

Geoscience and Exploration

G1 Geology and Mineral Exploration

G2 Applied Geophysics

G3 Applied Geochemistry

G4 Petroleum Geoscience

G5 Hydrogeology and Environmental Geoscience

Area B

Engineering and Technology

E1 Geological and Drilling Engineering

E2 Mining and Mineral Processing Engineering

E3 Petroleum Engineering

E4 Civil and Geotechnical Engineering

Area C

Space and Geospatial Technology

S1 Space Science and Technology

S2 Space Systems and Satellite Engineering

S3 Geomatics and Geospatial Engineering

Cross-disciplinary specialization: Geotourism, Geoheritage and Regional Development.

This gives you 13 professional major/specialization areas, but they do not all need to become separate bachelor's degree programs immediately.

I recommend starting with approximately eight or nine accredited degree programs, depending on available academic staff and laboratory capacity. The other areas can initially be offered as specialization tracks.

This distinction is important: **a faculty can have many research fields without having to establish an independent undergraduate program for every research field.**

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4. Recommended four-year educational curriculum and subject names

This is probably the most important part of the restructuring.

Your present problem is not simply that there are too many subjects. The more fundamental problem is that the curriculum does not clearly distinguish between:

1. General scientific foundations.
2. Common foundations of each academic discipline.
3. Professional subjects for each major.
4. Specialized and interdisciplinary subjects.

I recommend a curriculum with these four levels.

4.1 First level: common scientific and engineering foundation

All students should receive a strong scientific foundation, but the depth of individual subjects should depend on their major.

No.	Recommended subject	Summary
1	Calculus and Differential Equations	Differential and integral calculus, multivariable functions, vector calculus and differential equations used in scientific and engineering problems.
2	Linear Algebra and Numerical Methods	Matrix operations, eigenvalues, numerical solutions, computational mathematics and optimization.
3	Physics for Earth and Engineering	Mechanics, waves, electricity, magnetism, thermodynamics and the physical principles used in Earth and engineering sciences.
4	General and Analytical Chemistry	Chemical reactions, chemical equilibrium, solution chemistry, analytical methods and the chemical properties of natural materials.
5	Programming and Scientific Computing	Programming with Python and other relevant tools, numerical computation, algorithms, data processing and scientific visualization.

No.	Recommended subject	Summary
6	Probability, Statistics and Data Analytics	Probability theory, statistical inference, regression, uncertainty analysis and data interpretation.
7	Engineering Mechanics	Statics, dynamics, force systems, equilibrium and mechanical behavior of engineering systems.
8	Fluid Mechanics and Thermodynamics	Fluid properties, fluid flow, energy conservation, heat transfer and applications in groundwater, petroleum and hydraulic engineering.
9	Electrical and Electronic Engineering	Electrical circuits, electronic components, measurement systems, sensors and instrumentation.
10	Materials Science and Engineering	Physical and mechanical properties of materials, material testing, material selection and engineering applications.
11	Engineering Drawing and Computer-Aided Design	Technical drawing, engineering graphics, CAD, 3D modeling and basic engineering design.
12	Technical Communication and Professional Practice	Technical English, scientific writing, engineering reports, professional ethics, safety and teamwork.

An important recommendation: Do not make every one of these courses compulsory for every major at the same depth.

For example:

- Geology students need more chemistry, mineralogy and geochemistry.
- Geophysics students need more mathematics, physics, electronics and signal processing.
- Civil engineering students need more mechanics, materials science and fluid mechanics.
- Space engineering students need more physics, electronics, programming and control systems.
- Surveying students need more mathematics, statistics, programming and measurement theory.

This is how you can maintain a common scientific foundation without wasting educational time.

4.2 Second level: Geoscience and Exploration subjects



These subjects should form the principal academic foundation for students in geology, mineral exploration, geophysics and geochemistry.

No.	Recommended subject	Summary
1	Introduction to Earth System Science	Introduction to the structure, composition, origin and evolution of the Earth and the interactions between its geological, atmospheric, hydrological and biological systems.
2	Physical Geology	Introduction to geological materials, geological processes, geological structures and the physical evolution of the Earth.

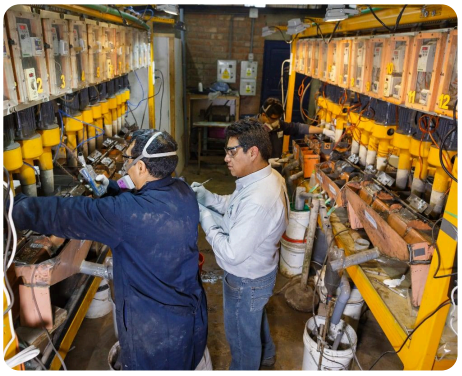
No.	Recommended subject	Summary
3	Crystallography and Mineralogy	Crystal structures, mineral classification, mineral identification, physical properties and chemical composition of minerals.
4	Igneous and Metamorphic Petrology	Origin, classification, composition and evolution of igneous and metamorphic rocks and their geological significance.
5	Sedimentology and Stratigraphy	Sedimentary processes, depositional environments, sedimentary structures, stratigraphic relationships and geological history.
6	Structural Geology and Tectonics	Geological deformation, faults, folds, fractures, tectonic processes and their relationships with mineral deposits and underground structures.
7	Geological Mapping and Field Methods	Geological observation, field measurement, geological mapping, cross-section preparation, field sampling and geological interpretation.
8	Economic Geology and Ore Deposits	Formation, classification, geological controls, mineral composition and distribution of metallic and non-metallic ore deposits.
9	Mineral Exploration and Resource Evaluation	Exploration planning, geological indicators, exploration methods, sampling, resource estimation, geological modeling and evaluation of mineral resources.
10	Applied Geophysics	Principles and applications of seismic, gravity, magnetic, electrical and electromagnetic methods for investigating underground geological structures.
11	Exploration Geochemistry	Geochemical sampling, analytical techniques, element distribution, geochemical anomalies and interpretation of mineralization indicators.
12	Hydrogeology and Groundwater Science	Groundwater occurrence, aquifer characteristics, groundwater movement, groundwater exploration, water quality and groundwater resource evaluation.
13	Petroleum Geology and Basin Analysis	Origin, migration and accumulation of hydrocarbons, sedimentary basin evolution, source rocks, reservoirs, seals and petroleum systems.
14	Geostatistics and 3D Geological Modeling	Statistical analysis of geological data, spatial correlation, interpolation, uncertainty assessment and three-dimensional geological modeling.

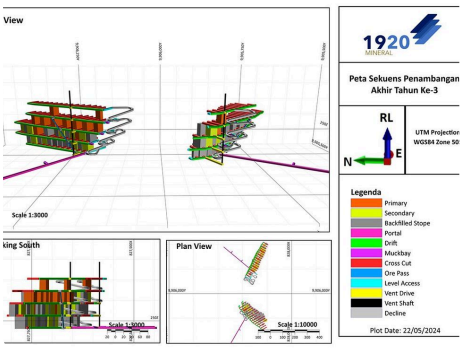
These subject selections are consistent with the scientific foundations of Oxford's Earth Sciences curriculum and the resource exploration programs at Jilin University. Both emphasize mineralogy, petrology, structural geology, geochemistry, geological fieldwork and quantitative methods. [Earth Sciences (Geology); 资源勘查工程-地球科学学院-吉林大学; 地质学-地球科学学院-吉林大学]

I would particularly recommend retaining Geological Mapping and Field Methods as a substantial practical course rather than treating fieldwork as a minor part of theoretical geology.

Cambridge and Oxford both emphasize field training and independent geological mapping as important components of Earth Sciences education. [Undergraduate fieldwork; Earth Sciences (Geology)]

4.3 Third level: Engineering and Technology subjects








These subjects should be divided into resource engineering and civil engineering.

A. Geological, drilling, mining and mineral engineering

No.	Recommended subject	Summary
1	Engineering Geology and Site Investigation	Application of geological knowledge to the investigation and evaluation of construction sites, foundations, slopes and underground structures.
2	Soil Mechanics and Foundation Engineering	Soil classification, stress distribution, consolidation, shear strength, bearing capacity and foundation design.
3	Rock Mechanics and Underground Engineering	Mechanical behavior of rock masses, rock deformation, underground excavation stability, slope stability and ground support.
4	Drilling Engineering and Well Design	Drilling principles, drilling methods, borehole design, drilling operations, drilling performance and well construction.
5	Drilling Equipment and Machinery	Structure, operation, selection, maintenance and development of drilling rigs, drill bits, pumps and related equipment.
6	Drilling Fluids and Well Control	Drilling fluid properties, fluid circulation, pressure control, borehole stability and prevention of drilling accidents.
7	Mine Design and Planning	Surface and underground mine design, mine layout, production planning, equipment selection and mine optimization.
8	Mining Engineering and Operations	Mining methods, extraction processes, mine development, mining machinery, ventilation, mine safety and production management.
9	Mineral Processing and Separation Engineering	Crushing, grinding, classification, gravity separation, magnetic separation, flotation, dewatering and mineral concentration.
10	Extractive Metallurgy and Mineral Materials	Extraction and recovery of valuable metals, hydrometallurgy, pyrometallurgy, recycling and development of mineral-based materials.
11	Mining Environmental Engineering	Mine drainage, tailings management, waste rock, environmental protection, mine rehabilitation and closure planning.
12	Mineral Resource Economics and Project Evaluation	Mineral resource economics, project feasibility, investment evaluation, operating costs, financial risk and resource management.

B. Petroleum engineering

No.	Recommended subject	Summary
13	Petroleum Engineering Fundamentals	Introduction to petroleum industry operations, exploration, development, drilling, production and reservoir management.
14	Petroleum Reservoir Engineering	Reservoir properties, fluid behavior, reservoir pressure, production mechanisms, recovery estimation and reservoir simulation.
15	Petroleum Production Engineering	Production systems, artificial lift, production optimization, well performance and production management.
16	Well Completion and Stimulation Engineering	Casing, cementing, perforation, completion equipment, hydraulic stimulation and well integrity.
17	Formation Evaluation and Well Logging	Interpretation of borehole measurements, formation properties, porosity, permeability, fluid saturation and reservoir evaluation.
18	Petroleum Processing and Surface Facilities	Separation of oil, gas and water, processing facilities, pipeline systems, storage, transportation and production equipment.
19	Petroleum Engineering Data Analytics	Statistical analysis, production data processing, reservoir data interpretation, simulation and digital oilfield applications.
20	Energy Resources and Sustainable Development	Petroleum resource development, geothermal energy, carbon capture and storage, energy economics and environmental responsibility.

Colorado School of Mines offers a useful model for these courses. Its petroleum curriculum includes drilling, completion, reservoir engineering, production, well logging, data analytics and project economics. Its mining curriculum includes mine design, rock mechanics and mineral processing. [Bachelor of Science in Petroleum Engineering; Bachelor of Science in Mining Engineering; Bachelor of Science in Mining Engineering]

C. Civil engineering









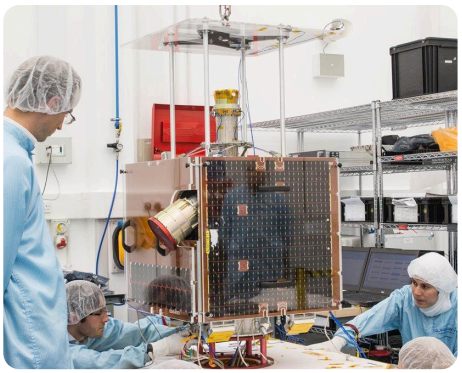



No.	Recommended subject	Summary
21	Statics and Mechanics of Materials	Equilibrium, internal forces, stress, strain, deformation and mechanical behavior of engineering structures.
22	Structural Analysis	Analysis of beams, frames, trusses and other structural systems under different loading conditions.

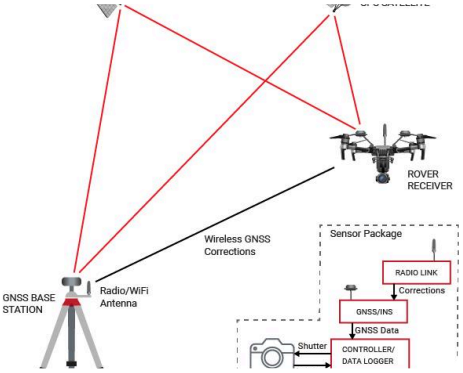
No.	Recommended subject	Summary
23	Reinforced Concrete and Steel Structures	Properties, design principles and structural behavior of reinforced concrete and steel structures.
24	Construction Materials and Technology	Properties, testing, production and application of concrete, steel, aggregates, rocks and other construction materials.
25	Foundation Engineering	Shallow and deep foundation design, settlement analysis, bearing capacity and foundation stability.
26	Geotechnical Engineering	Soil and rock behavior, earth structures, retaining walls, slope stability and geotechnical design.
27	Underground and Tunnel Engineering	Design and construction of tunnels, underground spaces, underground transport systems and excavation support.
28	Hydraulics and Water Resources Engineering	Hydraulic systems, open-channel flow, water supply, drainage, river engineering and water resource management.
29	Transportation and Infrastructure Engineering	Planning, design and construction of roads, bridges, transportation facilities and infrastructure networks.
30	Construction Engineering and Project Management	Construction methods, project scheduling, cost estimation, quality control, safety management and contract administration.
31	Building Information Modeling and Digital Construction	Application of BIM, digital modeling, construction information management and digital technologies in infrastructure development.

ETH Zurich, Tsinghua and the University of Tokyo offer useful references for structural engineering, geotechnical engineering, water resources, transportation, infrastructure and construction management. [Bachelor – Dept. of Civil, Environmental and Geomatic Engineering; Undergraduate-School of Civil Engineering Tsinghua University; (4) Undergraduate Departments]

4.4 Fourth level: Space Science and Geospatial Technology subjects







I recommend the following curriculum.

No.	Recommended subject	Summary
1	Fundamentals of Space Science	Introduction to the solar system, space environment, celestial bodies and fundamental physical processes in space.
2	Planetary Science and Comparative Planetology	Origin, composition, internal structure, geological processes and evolution of planets, moons and other planetary bodies.
3	Atmospheric Physics and Space Weather	Physical processes in the atmosphere, ionosphere, magnetosphere, solar wind and their effects on space activities.
4	Orbital Mechanics and Astrodynamics	Satellite orbits, orbital motion, orbital transfers, trajectory calculations and spacecraft motion.

No.	Recommended subject	Summary
5	Satellite Systems Engineering	Satellite architecture, spacecraft subsystems, power systems, communication, thermal control and mission requirements.
6	Spacecraft Sensors and Instrumentation	Design principles and applications of optical, radar, electromagnetic and other scientific instruments used in space missions.
7	Space Mission Design and Systems Engineering	Mission objectives, system requirements, mission architecture, spacecraft design, risk assessment and project development.
8	Earth Observation and Remote Sensing	Satellite observation principles, optical and radar sensors, image acquisition and monitoring of geological, environmental and surface processes.
9	Surveying, Geodesy and GNSS	Measurement principles, positioning, geodetic reference systems, GPS/GNSS, satellite navigation and surveying adjustment.
10	Photogrammetry, LiDAR and 3D Mapping	Image-based measurement, laser scanning, digital elevation models, point clouds and three-dimensional surface reconstruction.
11	Geographic Information Systems and Spatial Data Science	Spatial databases, GIS analysis, cartography, spatial modeling, geographic visualization and geospatial information management.
12	Space and Geospatial Data Analytics	Processing and interpretation of satellite and spatial datasets using statistical methods, machine learning and computational techniques.

These subjects reflect the educational directions of ETH Zurich, MIT and Peking University.

In particular, ETH's Space Systems program combines space systems engineering, space data, planetary science, space weather, Earth observation, aerospace engineering, robotics and space communication. MIT's Space Systems Engineering course also emphasizes integrated design of spacecraft subsystems and mission architecture. [Education - ETH Zurich; Department of Aeronautics and Astronautics; Master Space Systems]

4.5 My recommendation concerning the fundamental major subjects

I would make a significant change to your existing educational system.

At present, students across different detailed major indexes appear to study the same group of subjects, including mineralogy, petrology, physical geology and structural geology.

This is not necessarily appropriate.

For example:

Major	Essential common professional foundation
Geology and Mineral Exploration	Physical Geology, Mineralogy, Petrology, Structural Geology, Stratigraphy
Applied Geophysics	Mathematics, Physics, Geology, Geophysical Field Theory, Signal Processing
Applied Geochemistry	Chemistry, Mineralogy, Petrology, Geochemistry, Analytical Chemistry
Geological and Drilling Engineering	Engineering Geology, Mechanics, Fluid Mechanics, Rock Mechanics
Mining and Mineral Processing	Mineralogy, Petrology, Rock Mechanics, Mining Fundamentals, Materials Science
Petroleum Engineering	Petroleum Geology, Fluid Mechanics, Thermodynamics, Rock Properties, Engineering Mechanics
Civil and Geotechnical Engineering	Engineering Mechanics, Mechanics of Materials, Soil Mechanics, Engineering Geology
Space Science and Technology	Mathematics, Physics, Electromagnetism, Programming, Atmospheric and Space Physics
Geomatics and Geospatial Engineering	Mathematics, Statistics, Programming, Surveying, Geodesy, GIS

This table should become the basis of your curriculum reform.

The subjects that genuinely form a common foundation should be shared. Other subjects should be required only for the majors that need them.

5. Recommended synthetic and interdisciplinary subjects

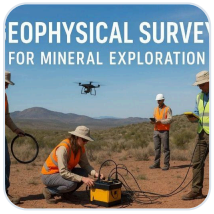
I consider this section especially important for your faculty.

One of your main advantages over a conventional geology faculty or civil engineering faculty is that you can educate students who understand several interconnected technical disciplines.

I recommend developing a group of synthetic subjects that combine geoscience, civil engineering and space/geospatial technologies.

These subjects should not simply repeat existing courses. They should teach students to solve practical problems by integrating the knowledge obtained from several departments.

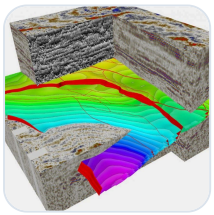
5.1 Proposed synthetic subjects



1. Integrated Mineral Exploration Engineering

High priority

Integrates geological mapping, economic geology, geophysics, geochemistry, drilling, remote sensing and GIS. Students develop a complete exploration plan, identify prospective areas, select exploration methods and evaluate mineral resources.



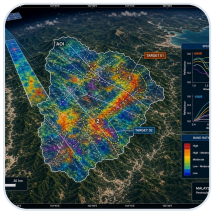
2. Integrated Subsurface Imaging and 3D Geological Modeling

Combines geological interpretation, seismic data, gravity and magnetic data, geochemical information, borehole measurements and computational modeling to develop three-dimensional and four-dimensional subsurface models.



3. Geological Engineering for Civil Infrastructure

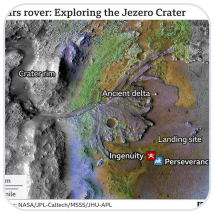
Integrates structural geology, engineering geology, soil mechanics, rock mechanics and civil engineering design. Students evaluate geological conditions and develop solutions for foundations, tunnels, dams, bridges and other infrastructure.



4. Satellite Remote Sensing for Mineral and Infrastructure Exploration

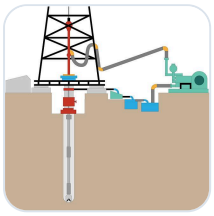
Combines satellite imagery, multispectral and hyperspectral analysis, radar remote sensing, GIS and geological interpretation for mineral exploration,

construction site selection, environmental monitoring and infrastructure assessment.



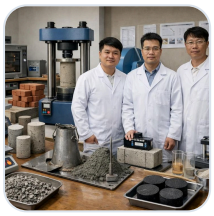
5. Planetary Geology and Space Resource Exploration

Integrates planetary science, geology, geophysics, geochemistry, remote sensing and mineral resource evaluation to investigate planetary surfaces, geological evolution and potential extraterrestrial resources.



6. Integrated Drilling and Subsurface Engineering

Combines geological investigation, drilling technology, rock mechanics, drilling fluids, well logging, petroleum engineering and geotechnical engineering to develop drilling systems and subsurface investigation plans.



7. Mineral Materials and Sustainable Construction Technology

Combines mineralogy, mineral processing, extractive metallurgy, materials science and civil engineering to develop construction materials and other industrial products from minerals and rocks.



8. Geospatial Engineering and Smart Infrastructure

Combines surveying, GNSS, photogrammetry, GIS, remote sensing, civil engineering and digital technologies to develop digital maps, infrastructure monitoring systems and digital twins.



9. Geoheritage, Geotourism and Sustainable Regional Development

Integrates geology, geomorphology, GIS, environmental science, civil infrastructure and tourism planning to evaluate geological attractions and develop sustainable tourist districts and geoparks.



10. Environmental and Social Management of Resource Development

Combines geology, mining engineering, hydrogeology, civil engineering, environmental assessment, safety, resource economics and social responsibility for sustainable resource development.



11. Earth Observation and Space Mission Design

Combines space science, orbital mechanics, satellite instrumentation, remote sensing, communications and geospatial data analysis. Students develop mission concepts for monitoring geological, atmospheric and environmental phenomena.



12. Integrated Earth Resources and Infrastructure Design Project

A senior-level project in which students from different majors cooperate to develop a resource exploration and development project, including geological assessment, geophysical surveys, drilling, mining or petroleum engineering, infrastructure design, economic evaluation and environmental management.

5.2 How should these synthetic subjects be introduced?

I would divide them into three categories.

Category	Subjects	Recommended students
Resource exploration integration	1, 2, 4, 6	Geology, geophysics, geochemistry, drilling and petroleum students
Engineering integration	3, 6, 7, 10, 12	Civil, geological, mining, mineral and petroleum engineering students
Space and geospatial integration	4, 5, 8, 9, 11	Space science, surveying, GIS, remote sensing and regional development students

Students should not be required to take all twelve subjects.

Instead, I recommend:

- One or two integrated subjects as compulsory requirements within each major.
- A choice of interdisciplinary subjects from related departments.
- One multidisciplinary senior design project during the fourth year.

This will help reduce repeated theoretical content and increase students' ability to solve real-world problems.

6. Recommended four-year curriculum structure

Your university has a four-year academic system. I recommend dividing the educational process into four stages.

The main principle should be:

Broad scientific foundation → disciplinary foundation → professional specialization → integrated engineering and research practice.

This approach is consistent with the educational structures of Peking University, ETH Zurich, Oxford and other universities, although their degree durations and credit systems differ. Peking University, for example, uses a four-year undergraduate system with common foundational education followed by specialization. ETH and Oxford emphasize foundational study before advanced specialization and independent projects. [培养方案-北京大学地球与空间科学学院; Bachelor in Earth and Climate Sciences – Department of Earth and Planetary Sciences; Earth Sciences (Geology)]

Proposed curriculum by academic year

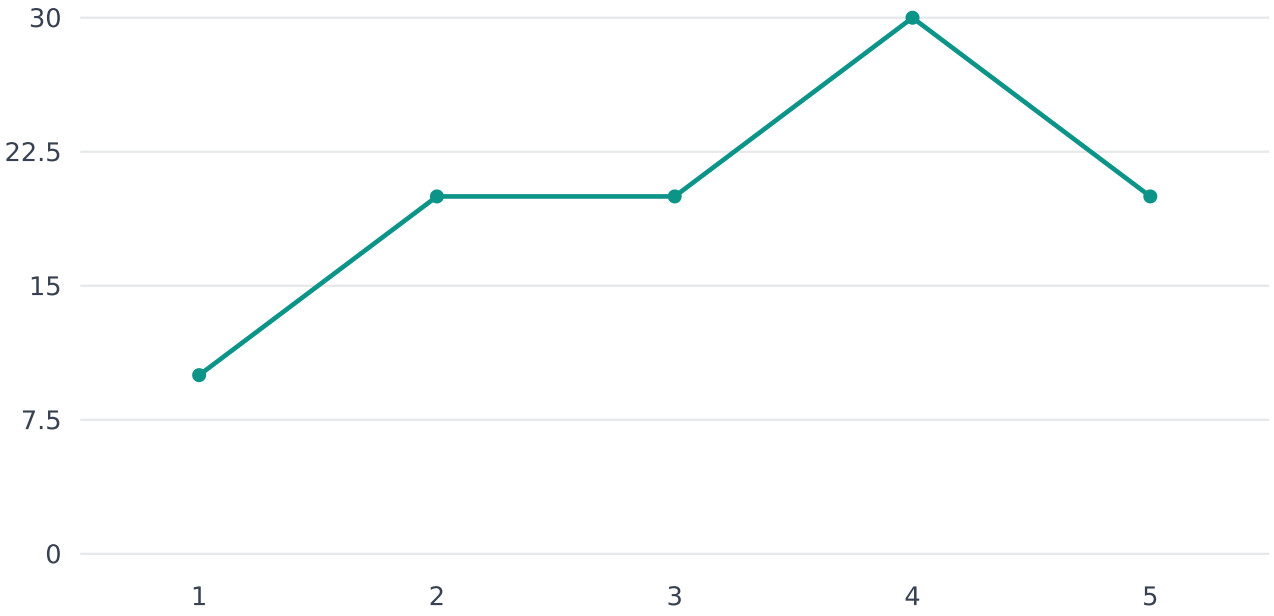
Academic year	Educational objective	Main educational content
First year	Scientific foundation	Mathematics, physics, chemistry, programming, engineering fundamentals, introduction to Earth science, engineering and space science.
Second year	Disciplinary foundation	Fundamental major subjects, laboratory experiments, field methods, engineering mechanics and introductory professional subjects.
Third year	Professional specialization	Advanced major subjects, specialized laboratory work, field surveys, engineering design and interdisciplinary subjects.
Fourth year	Professional competence and independent work	Industrial internship, integrated design projects, graduation thesis, advanced electives and professional practice.

Recommended curriculum distribution

As an initial planning model, I suggest the following distribution of total graduation credits:

Recommended distribution of graduation credits

An indicative curriculum allocation, to be adjusted to your university's credit regulations.



Credit allocation

The practical component should include:

- Geological field mapping.
- Geophysical and geochemical exploration exercises.
- Drilling and mining practice.
- Mineral processing laboratory practice.
- Civil construction and geotechnical site investigation.
- Surveying and remote sensing fieldwork.
- Satellite data analysis and space-related projects.
- Industrial internships.

I would especially emphasize practical training in the third and fourth years.

For geology students, a substantial geological mapping project should be compulsory. For engineering students, the main requirement should be an engineering design project. For space and geospatial students, the main requirement should be a satellite-data or space-systems project.

Example of a fourth-year integrated project

A particularly valuable project would be:

Integrated Mineral Resource Exploration and Development Project

Students from different majors work together:

Student specialization	Responsibility
Geology	Geological mapping and interpretation
Geophysics	Geophysical survey design and interpretation
Geochemistry	Geochemical sampling and anomaly evaluation
Drilling engineering	Drilling design and exploration drilling plan
Mining engineering	Mine design and production planning
Mineral processing	Mineral beneficiation and processing design
Civil engineering	Infrastructure and construction planning
Geospatial engineering	GIS, satellite data and 3D modeling

Students jointly prepare a technical report, economic evaluation, engineering design and environmental management plan.

This kind of project would distinguish your faculty from a conventional theoretical geoscience faculty.

7. Additional strategic recommendations for the Dean

Beyond the organization and curriculum, I would suggest several important strategic initiatives.

7.1 Establish a clear educational philosophy

I recommend the following educational philosophy:

To cultivate internationally competitive geoscientists and engineers who possess strong scientific foundations, advanced technological knowledge,

practical engineering capabilities and interdisciplinary problem-solving skills.

The faculty should educate students who can:

1. Understand geological processes and the formation of mineral and energy resources.
2. Discover and evaluate mineral and petroleum resources.
3. Apply geophysical and geochemical technologies to resource exploration.
4. Design and operate drilling, mining and mineral processing systems.
5. Investigate geological conditions and design civil infrastructure.
6. Develop and apply satellite, remote sensing and geospatial technologies.
7. Manage engineering projects economically, safely and sustainably.

7.2 Establish three major research and educational directions

I recommend concentrating the faculty's research activities into three strategic directions.

Strategic direction 1

Earth Resources Exploration and Evaluation

- Critical and strategic mineral exploration.
- Geological and geophysical exploration.
- Exploration geochemistry.
- Petroleum and natural gas exploration.
- Hydrogeology and groundwater.
- Deep geological exploration.

Strategic direction 2

Earth Resources Engineering and Infrastructure Development

- Drilling and subsurface engineering.
- Mining and mineral processing.

- Petroleum engineering.
- Civil and geotechnical engineering.
- Mineral materials and industrial applications.
- Sustainable infrastructure development.

Strategic direction 3

Space Science, Earth Observation and Geospatial Technology

- Space science and planetary exploration.
- Satellite systems and instrumentation.
- Atmospheric and space environment research.
- Remote sensing and Earth observation.
- Surveying and geospatial information.
- Geospatial applications in regional development.

This strategic arrangement will also help you decide which research equipment to purchase, which academic staff to recruit, and which international universities to cooperate with.

7.3 Strengthen digital and intelligent technologies

I would recommend introducing the following technologies across the faculty:

- Artificial intelligence and machine learning.
- Geological big-data analysis.
- 3D and 4D geological modeling.
- Automated geophysical interpretation.
- Digital drilling systems.
- Intelligent mining equipment.
- Digital twins for civil infrastructure.
- UAV and satellite remote sensing.
- GIS and geospatial intelligence.

These should not be confined to one laboratory or one major.

For example, AI and programming should be introduced in the common scientific foundation, while specialized applications should be taught within each department.

7.4 Establish international educational cooperation

I recommend establishing academic cooperation with universities that have complementary strengths.

University	Suggested cooperation area
Colorado School of Mines	Mineral exploration, geological engineering, mining and petroleum engineering
ETH Zurich	Geoscience, geophysics, geospatial engineering and space systems
Peking University	Geology, geochemistry, geophysics and space science
Jilin University	Exploration technology, drilling, surveying and geological engineering
Tsinghua University	Civil engineering, construction management and aerospace engineering
MIT	Applied geoscience, civil engineering, space systems and computational methods
University of Cambridge	Fundamental geoscience, field geology and Earth systems
University of Tokyo	Resource engineering, civil engineering, Earth sciences and aerospace

I would prioritize Colorado School of Mines and Jilin University for your resource exploration and engineering programs, ETH Zurich for space and geospatial technologies, and Tsinghua for civil engineering.

8. Implementation roadmap

I would not recommend changing all departments and curricula simultaneously. A gradual transition would reduce organizational confusion and educational disruption.

Phase	Period	Main activities
Phase 1: Academic diagnosis	First 3–6 months	Review existing staff, laboratories, research fields, courses, duplicated subjects and industry needs.
Phase 2: Organizational reform	6–12 months	Establish the five departments, clarify academic responsibilities and reorganize laboratory management.
Phase 3: Curriculum reform	1–2 years	Develop new major programs, course descriptions, learning outcomes and integrated subjects.
Phase 4: Educational implementation	2–3 years	Introduce the new curriculum, practical training, industrial internships and integrated graduation projects.
Phase 5: International development	3–5 years	Develop international cooperation, external accreditation, joint research and continuous curriculum improvement.

Before approving the new curriculum, I would also conduct a detailed audit of existing courses.

For every subject, the faculty should identify:

- Which department owns the subject?
- Which majors require it?
- What knowledge and skills should students acquire?
- Which other subjects have overlapping content?
- What practical training is required?
- What are its prerequisites?
- Is the subject still relevant to current industry requirements?

I recommend maintaining a curriculum mapping matrix for every major. This will help prevent the problem of increasing elective subjects and repeated content that you identified in your initial assessment.

9. Final recommendations and conclusions

After considering the international university structures, your existing academic organization, your four-year educational system and your faculty's development objectives, my recommendations are as follows.

Your question	My recommendation
1. Is the current academic organization suitable?	The integrated faculty concept is suitable, but its internal academic responsibilities and boundaries need substantial improvement.
2. What academic organization should be adopted?	Establish five departments: Earth Sciences and Mineral Exploration; Applied Geophysics and Geochemistry; Mineral, Mining, Petroleum and Drilling Engineering; Civil and Geological Engineering; Space Science and Geospatial Technology.
3. What detailed major indexes should be introduced?	Develop nine principal degree programs initially, with additional specializations in hydrogeology, petroleum geology, satellite systems, geotourism and other fields according to staff capacity.
4. What subjects should be introduced?	Establish a common scientific foundation, separate disciplinary foundations, professional major subjects and practical engineering subjects.
5. What synthetic subjects should be introduced?	Establish integrated exploration, subsurface modeling, geological engineering, satellite remote sensing, planetary resources, drilling, sustainable materials and interdisciplinary graduation projects.

My five most important strategic decisions for you

If I were responsible for leading this faculty reform, I would make the following five decisions.

First: Maintain the integrated science-and-engineering identity.

Do not divide your faculty into a theoretical geology faculty and a separate engineering faculty. Your integrated structure is a strategic advantage, especially for a university that aims to develop highly qualified specialists in mineral exploration, petroleum, civil engineering and space technology.

Second: Establish five departments with clear academic responsibilities.

The five-department structure should become the foundation of your academic organization. Departments should own degree programs and educational content, while laboratories should support interdisciplinary research and practical education.

Third: Reorganize the major indexes around internationally recognized professional disciplines.

Geology, geophysics, geochemistry, geological engineering, drilling engineering, mining engineering, mineral processing, petroleum engineering, civil engineering, space science and geomatics should have clear educational identities.

Fourth: Reform the curriculum to eliminate unnecessary overlap.

The most important change is to distinguish common scientific subjects from professional foundation subjects and specialized subjects. Do not force every major to study the same theoretical geology subjects.

Fifth: Make interdisciplinary engineering education a distinguishing characteristic of the faculty.

The integrated subjects and graduation projects should be among the faculty's strongest educational features.

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10. A proposed long-term identity for the faculty

I would summarize the future direction of your faculty with the following concept:

Faculty of Geoscience and Technology

Earth · Resources · Infrastructure · Space

Scientific understanding and technological development of the Earth and space

Three interconnected educational missions:



Earth Science



Engineering



Space Technology

Discover and understand Earth resources.	Develop resources and infrastructure.	Explore space and utilize geospatial information.

Strong Science + Advanced Technology + Practical Engineering + Interdisciplinary Education

One final thought: I would not measure the success of this reform by the number of departments, laboratories or major programs established.

I would measure it by whether graduates can independently solve complex professional problems.

A graduate of your faculty should be capable of understanding a geological problem, selecting appropriate exploration methods, processing geological and geophysical information, designing an engineering solution, evaluating its economic feasibility and considering its environmental consequences.

That combination of abilities could become the distinctive identity of the Faculty of Geoscience and Technology.

My strongest recommendation is to build the faculty as an integrated Earth Science and Engineering institution, with Space Science and Geospatial Technology as its third strategic pillar. This would distinguish it from conventional geology faculties and conventional engineering faculties while maintaining a clear academic structure.

I have treated this as a strategic academic restructuring proposal based on published university structures and curricula, rather than a completed institutional feasibility study. Before formal adoption, the proposal should be checked against your existing staff qualifications, student enrollment, laboratory capacity, national higher-education regulations and local industry requirements.

A particularly useful next step would be to develop a **complete four-year, eight-semester curriculum for each of the nine proposed majors**, including compulsory and elective subjects, credit allocation, prerequisites, laboratory hours, fieldwork and graduation requirements. That would turn this strategic proposal into a document that your Faculty Academic Council could evaluate and implement.

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